

Perpendicular lines and planes in space

Two lines is enough — the test that turns an infinite condition into a finite check.

LESSON 33–34

2 × 40 MIN

GEOMETRY 10, §14

EXTENSION

LESSON CLOCK

12'

24'

22'

20'

16'

6'

The definition	12 min
The test	24 min
Two lemmas	22 min
Construction and uniqueness	20 min
Practice	16 min
Homework	6 min

LEARNING OBJECTIVES

- Define a line perpendicular to a plane.
- State and apply the test for perpendicularity of a line and a plane.
- Prove the two lemmas on parallel lines and perpendicular planes.
- Construct the perpendicular from a point to a plane.

TEXTBOOK

National	pp. 147–158
Cambridge	Core & Extended, pp. 323–328
Workbook	Exercise 14.1

01

Explanation

The definition

DEFINITION

A line is **perpendicular to a plane** when it is perpendicular to **every** line lying in that plane.

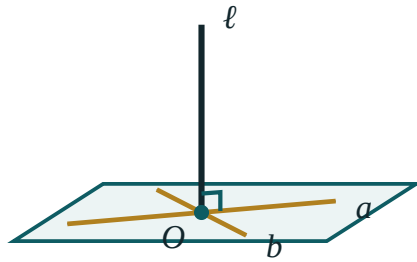
As a definition that is unusable — there are infinitely many lines to check. The value of the next theorem is that it reduces “every” to “two”.

The line is also called a **normal** to the plane, and the point where it meets the plane is the **foot** of the perpendicular.

The test

TEST FOR A LINE PERPENDICULAR TO A PLANE

If a line is perpendicular to **two intersecting** lines of a plane, it is perpendicular to the whole plane.



$\ell \perp a$ and $\ell \perp b$, with a and b intersecting at O
therefore ℓ is perpendicular to the whole plane

Two intersecting lines are enough. A third would be redundant; one would not be enough.

INTERSECTING, NOT PARALLEL

A line perpendicular to two **parallel** lines of a plane need not be perpendicular to the plane: a pencil laid across two rails of a railway is perpendicular to both, and lies flat in the plane. The two directions must be genuinely different.

In a cube, $AA_1 \perp AB$ and $AA_1 \perp AD$, and AB and AD intersect at A . So $AA_1 \perp ABCD$, and therefore $AA_1 \perp AC$ and $AA_1 \perp BD$ as well — results that would otherwise each need their own proof.

Two lemmas

Lemma 1. If one of two parallel lines is perpendicular to a plane, so is the other.

Lemma 2. If two lines are perpendicular to the same plane, they are parallel.

Together these say: **all the normals of a plane are parallel to one another**, and any line parallel to a normal is itself a normal. The direction perpendicular to a plane is unique.

$$a \perp \alpha \text{ and } b \perp \alpha \Rightarrow a \parallel b$$

COMPARE WITH THE PLANE CASE

In the plane, two lines perpendicular to the same line are parallel — the same statement, one dimension down. Lemma 2 is its space version, and it is proved the same way.

Construction and uniqueness

Theorem. Through any point there passes exactly one line perpendicular to a given plane, and exactly one plane perpendicular to a given line.

The distance from a point to a plane is the length of that perpendicular segment — and by the argument of the next lesson, it is the shortest of all segments from the point to the plane.

IN A CUBE $ABCD A_1 B_1 C_1 D_1$	PERPENDICULAR TO
AA_1	$ABCD$ and $A_1 B_1 C_1 D_1$
AB	$ADD_1 A_1$ and $BCC_1 B_1$
AD	$ABB_1 A_1$ and $DCC_1 D_1$

02 Worked examples

EXAMPLE 1 In a cube, prove $AA_1 \perp AC$.

- ① $AA_1 \perp AB$ and $AA_1 \perp AD$.
Adjacent edges of square faces.
- ② AB and AD intersect at A and lie in $ABCD$.
- ③ By the test, $AA_1 \perp ABCD$.
- ④ $AC \subset ABCD$.
By the definition.

ANSWER

$AA_1 \perp AC$

EXAMPLE 2 A line is perpendicular to two parallel lines of a plane. Must it be perpendicular to the plane?

- 1 The test demands two **intersecting** lines.
- 2 A line inside the plane, crossing both, is perpendicular to both.
- 3 It is not perpendicular to the plane.

ANSWER

No

EXAMPLE 3 Two lines a and b are perpendicular to the plane α . What can be said?

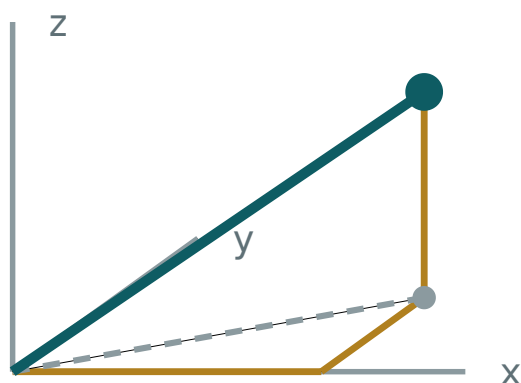
- 1 Lemma 2 applies.
- 2 $a \parallel b$
- 3 They determine a plane, perpendicular to α .

ANSWER

 $a \parallel b$ **03** Interactive model

Stand a pencil on the desk and check it against two rulers laid across it.

A normal to a plane



x 3

y 2

z 2

OM 4.12

base diagonal 3.61

04 Terminology

Write these on the board at the start. Learners meet the national programme in Uzbek or Russian and the Cambridge papers in English — the three columns are the same idea.

ENGLISH	O'ZBEKCHA	РУССКИЙ
Perpendicular to a plane	Tekislikka perpendikulyar	Перпендикуляр к плоскости
Test for perpendicularity	Perpendikulyarlik alomati	Признак перпендикулярности
Two intersecting lines	Kesishuvchi ikki chiziq	Две пересекающиеся прямые
Foot of the perpendicular	Perpendikulyar asosi	Основание перпендикуляра
Normal to a plane	Tekislik normali	Нормаль к плоскости
Unique perpendicular	Yagona perpendikulyar	Единственный перпендикуляр
Lemma	Lemma	Лемма
Construction	Yasash	Построение

Stress the words in bold in the explanation above; ask for the Uzbek and Russian back before moving on.

05 Quick check**Check yourself**

A line $\perp$ to a plane is $\perp$ to:

one line of it

two lines of it

every line of it

no line of it

The test needs:

one line

two intersecting lines

two parallel lines

three lines

Two lines $\perp$ to the same plane are:

perpendicular

parallel

skew

equal

If $a \parallel b$ and $a \perp \alpha$ then:

$$b \parallel \alpha$$

$$b \perp \alpha$$

$$b \subset \alpha$$

nothing

In a cube AA_1 is perpendicular to:

$$ABB_1A_1$$

$$ABCD$$

$$ADD_1A_1$$

all of them

06 Practice · 21 problems

Seven at each level. Set the easy row for everyone, the medium row for the main body of the class, and the hard row for those who finish early.

Easy · warm the idea up

7 PROBLEMS

1. Define a line perpendicular to a plane

ANSWER $\perp$ to every line of the plane

2. State the test

ANSWER $\perp$ to two intersecting lines of the plane

3. Is $AA_1 \perp ABCD$ in a cube?

ANSWER yes

4. Is $AA_1 \perp AC$?

ANSWER yes

5. Two lines $\perp$ to the same plane are

ANSWER parallel

6. Is one line enough for the test?

ANSWER no

7. Are two parallel lines enough?

ANSWER no

Medium · the standard question

7 PROBLEMS

1. Prove $AA_1 \perp BD$ in a cube

ANSWER $AA_1 \perp ABCD$ and $BD \subset ABCD$

2. Which faces is AB perpendicular to?

ANSWER ADD_1A_1 and BCC_1B_1

3. If $a \perp \alpha$ and $b \perp \alpha$, what is the angle between a and b ?

ANSWER 0° — parallel

4. How many perpendiculars to a plane through a point?

ANSWER 1

5. How many planes $\perp$ to a line through a point?

ANSWER 1

6. Give a counterexample to "two parallel lines suffice"

ANSWER A line inside the plane crossing two rails

7. In a cube, is $AC \perp BD$?

ANSWER yes — diagonals of a square

Hard · stretch and reason

7 PROBLEMS

1. Prove Lemma 1

ANSWER Translate the perpendicularity along the parallel

2. Prove Lemma 2

ANSWER Otherwise two planes through one point would be perpendicular to one line

3. In a cube, prove $BD \perp AA_1C_1C$ ANSWER $BD \perp AC$ and $BD \perp AA_1$ 4. In a cube, prove $AC_1 \perp BD$ ANSWER $BD \perp AA_1C_1C$ contains AC_1 5. A regular pyramid: prove the axis is $\perp$ to the baseANSWER It is $\perp$ to two base diagonals6. If $a \perp \alpha$ and $\alpha \parallel \beta$, is $a \perp \beta$?

ANSWER yes

7. Can a line be perpendicular to two intersecting planes?

ANSWER no — the normals would be parallel

07 Homework**Homework — 5 tasks**

Every proof must name the two intersecting lines used in the test.

1. In cube $ABCD A_1 B_1 C_1 D_1$, prove $DD_1 \perp AC$.
2. List every face of a cube that BC is perpendicular to.
3. Prove that if two lines are perpendicular to the same plane they are parallel.
4. Give a counterexample showing that two parallel lines are not enough for the test.
5. In a cube, prove $BD \perp AA_1 C_1 C$ and deduce $BD \perp AC_1$.